Hall effect in high-purity and p-doped germanium crystals*

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Experimental and theoretical investigations of the Hall effect were carried out in high-purity (10^{10}-cm^{-3} carrier concentration) germanium crystals. Optical phonon scattering, which was not included in the previous work on the calculation of the Hall coefficient factor $\hat{\theta}$, is included in the present calculation. Extensive numerical calculations were performed for $\hat{\theta}$ vs $1/T$ for both the spherical-band model and the anisotropic model of Beer and Willardson. Measurements and calculations of the Hall coefficient vs $1/T$ are in excellent agreement.

I. INTRODUCTION

The Hall effect in p-type germanium crystals has been investigated both theoretically and experimentally.[1-7] In particular, Willardson and co-workers have made a detailed study of the transverse Hall and magnetoresistance effect as a function of magnetic field. Beer and Willardson[6] found that when anisotropic effects are included, calculations of the Hall effect agree very well with actual measurements.

In all previous investigations of the Hall effect, the doping level has been on the order of 10^{13} cm^{-3} or higher. Furthermore, crystal uniformity has not been very good, and there have always been unknown impurities. As a result, certain fine effects, which might otherwise be revealed in the Hall-effect experiment, have been obscured by the presence of these unknown impurities. Hall-effect measurements in undoped crystals in the 10^{10}—10^{12}-cm^{-3} impurity concentration range have not been investigated in the past because such pure crystals were not available.

Recently, due to the demand for high-purity germanium crystals for γ-ray detectors, extensive effort has been put into producing purified germanium. As a result, germanium crystals with impurity concentrations in the 10^{10}—10^{9}-cm^{-3} range are now available.

Extensive measurements of Hall effect versus temperature in the 10^{10}—10^{12}-cm^{-3} purity range were carried out by Hall *et al.*[7] The effect of the low-level radiation damage on these high-purity germanium crystals was studied by Blankenship.[8]

One of the more interesting questions that arose from the work of Hall *et al.*[7] with high-purity germanium concerns the change of the slope in the Hall—effect-vs-temperature curve in the range from about 80 °K to the temperature at which the onset of intrinsic conduction occurs. The change of the slope had also been observed at the Lawrence Livermore Laboratory in previous Hall measurements of more heavily doped samples (up to 4×10^{15} cm^{-3}). Although Hall *et al.* were the first to give the correct interpretation of this phenomenon, they did not carry out numerical calculations for comparison with their experimental results.

Likewise, Blankenship[8] did not perform theoretical calculations for the Hall coefficient factor. In fact, in the work of Blankenship the influence of the magnetic field on the Hall effect was completely neglected for the Hall-effect measurement, both theoretically and experimentally.

The purpose of this paper is (i) to explain the change of the slope in the Hall—effect-vs-temperature curve and (ii) to expand the work of Beer and Willardson,[6] who while having carried out extensive numerical calculations on the Hall-effect coefficient factor, have not included optical phonon scattering in their calculation. Furthermore, their calculation was strictly for $\hat{\theta}$ as a function of magnetic field—rather than $\hat{\theta}$ vs temperature, as has been done in this paper.

II. THEORY

For the numerical calculations for the Hall coefficient we shall use both the spherical-band approximations and the Beer-Willardson models.[6] The latter takes into consideration the anisotropic effect of the heavy valence band, and treats the light band as being spherical. With these approximations in mind we shall now derive the Hall coefficient factor $\hat{\theta}$ when optical phonon scattering is taken into consideration.

We take the relaxation time for the acoustic phonon to be of the form[5]

$$\tau_{ai}=\bar{A}_i T^{-1}x^{-1/2}=\tau_{Li}x^{-1/2}, \tag{1}$$

where T is the temperature, $i=1$ for the conduction band, $i=2$ for the heavy-hole band, $i=3$ for the light-hole band, x is the energy of the carriers divided by kT, and $\bar{A}_i$ is a constant that contains the deformation potential as determined by fitting the experimental values of the temperature dependence of the mobility.

Following Conwell,[9] the reciprocal relaxation time for scattering by the optical modes, which includes emission of an optical phonon and absorption of an optical phonon, can be written

$$\frac{1}{\tau_{0i}}=\frac{1}{\tau_{Li}}\frac{B}{A}\frac{\theta}{T}\left[\frac{1}{[\exp(\theta/T)-1]}\left(1+\frac{k\theta}{E}\right)^{1/2}+\left(\frac{1}{\exp(\theta/T)-1}+1\right)\left(1-\frac{k\theta}{E}\right)^{1/2}\right]=\frac{1}{\tau_{Li}}\alpha(x) \tag{2}$$

where

$$\alpha(x)=\frac{B}{A}\frac{\theta}{T}n(\theta)\left[\left(x+\frac{\theta}{T}\right)^{1/2}+\exp\left(\frac{\theta}{T}\right)\left(x-\frac{\theta}{T}\right)^{1/2}\right],$$

where $k\theta$ is the optical phonon energy, $n(\theta)=[\exp(\theta/T)-1]^{-1}$, and $x=E/kT$. In using Eq. (2), the optical phonon energy is taken to be 0.038 eV.[5] The ratio of B/A is equal to $(\epsilon_{op})^2/2(\epsilon_{ac})^2$, where ϵ_{op} and ϵ_{ac} are the deformation potentials of the valence bands.

For the impurity scattering we shall use the Brooks-Herring formula for the light and heavy holes, e.g.,[6,10]

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$$\tau_{Ii} = \mu_{Ii}\tfrac{1}{8}\pi^{1/2}x^{3/2}m_i^*/e, \tag{3}$$

$$\mu_{Ii} = 2^{7/2}\pi^{-3/2}(kT)^{3/2}\epsilon^2 e^{-3} \times (m_i^*)^{-1/2}(N_I)^{-1}\{\ln[6m_i(kT)^2\epsilon/\pi e^2\hbar^2 p']\}^{-1}, \tag{4}$$

$$p' \equiv p + (p+N_D)[1-(p+N_D)/N_A],$$

$$N_I \approx p + 2N_D.$$

N_I is equal to the total concentration of ionized centers, N_D and N_A are the concentrations of ionized donor and acceptor centers, respectively, p is the total hole concentration, and ϵ is the dielectric constant.

Combining Eqs. (1)—(3), the total relaxation can be written

$$\frac{1}{\tau_{Ti}} = \frac{1}{\tau_{0i}} + \frac{1}{\tau_{ai}} + \frac{1}{\tau_{Ii}}, \tag{5}$$

$$\tau_{Ti} = \frac{3\pi^{1/2}}{4}\frac{m_i^*}{e}\frac{\mu_{Li}x^{3/2}}{x^2+\bar{\beta}_i(x)}, \tag{6}$$

$$\bar{\beta}_i(x) = 6\frac{\mu_{Li}}{\mu_{Ii}} + x^{3/2}\alpha(x), \tag{7}$$

where

$$\mu_{Li} = \tfrac{4}{3}(1/\sqrt{\pi})(e/m_i^*)\tau_{Li}.$$

For the derivation of the Hall coefficient factor $\hat{\theta}$ we need the expression for the conductivity coefficients S_{xx} and S_{xy}. Based upon the theory of McClure, [11] Beer and Willardson show that these two coefficients can be written in the following forms[6]:

$$S_{xx} = \frac{2}{3}\frac{pe^2}{m^*}\alpha\left\langle x\tau\sum_{N\ \mathrm{odd}}\frac{b_N}{1+(N\omega\tau)^2}\right\rangle, \tag{8}$$

$$S_{xy} = \frac{2}{3}\frac{pe^2}{m^*}\alpha\left\langle x\omega\tau^2\sum_{N\ \mathrm{odd}}\frac{b_N(-1)^n N}{1+(N\omega\tau)^2}\right\rangle, \qquad n \equiv \tfrac{1}{2}(N-1) \tag{9}$$

where the average over energy is defined by

$$\langle q\rangle \equiv (2/\pi^{1/2})\int_0^\infty x^{1/2}e^{-x}q\,dx. \tag{10}$$

The Hall coefficient is given by

$$R_H = S_{xy}[H(S_{xx}^2+S_{xy}^2)]^{-1}. \tag{11}$$

By definition the Hall coefficient factor is

$$\hat{\theta} \equiv R_H/R_\infty, \tag{12}$$

where R_∞ is obtained from Eq. (11) by letting the magnetic field approach infinity, i.e.,

$$R_\infty = \lim_{H\to\infty} R_H. \tag{13}$$

By substituting the expression for the total relaxation given in Eq. (5) into Eqs. (8), (9), (11), and (12), it can be shown that, after some simple algebra, Eq. (13) can be written

$$R_\infty = 1/e(p_2+p_3), \tag{14}$$

where p_2 and p_3 are the contributions from the heavy- and light-hole bands, respectively, and the Hall coefficient factor is

$$\hat{\theta} = \frac{R_H}{R_\infty} = \frac{3\pi}{8}\left(1+\frac{p_3}{p_2}\right)\frac{\mathcal{L}(\beta_2,\gamma_2)+(p_3/p_2)(\mu_{L,3}/\mu_{L,2})^2 L(\beta_3,\gamma_3)}{[K(\beta_2,\gamma_2)+(p_3/p_2)(\mu_{L,3}/\mu_{L,2})K(\beta_3,\gamma_3)]^2+\frac{1}{4}\pi\gamma_2[\mathcal{L}(\beta_2,\gamma_2)+(p_3/p_2)(\mu_{L,3}/\mu_{L,2})^2 L(\beta_3,\gamma_3)]^2}, \tag{15}$$

where

$$\mathcal{L}(\beta_2,\gamma_2) \equiv a\alpha[L(\beta_2,a^2\gamma_2) - 3b_3L(\beta_2,9a^2\gamma_2) + 5b_5L(\beta_2,25a^2\gamma_2)]$$

and

$$K(\beta_2,\gamma_2) \equiv \alpha[K(\beta_2,a^2\gamma_2) + b_3K(\beta_2,9a^2\gamma_2) + b_5K(\beta_2,25a^2\gamma_2)].$$

Now,

$$L(\beta_i,C\gamma_i) = \frac{2}{\pi^{1/2}}\int_0^{\theta/T}\frac{x^{9/2}e^{-x}\,dx}{[x^2+\beta_i^1(x)]^2+C\gamma_i x^3} + \int_{\theta/T}^\infty \frac{x^{9/2}e^{-x}\,dx}{[x^2+\beta_i(x)]^2+C\gamma_i x^3},$$

where

$$\beta_i^1(x) = \beta_i + \left[\frac{A}{B}\frac{\theta}{T}\left(\frac{1}{\exp(\theta/T)-1}\right)\left(x+\frac{\theta}{T}\right)^{1/2}\right]x^{3/2}, \qquad \theta > T, \tag{16}$$

$$\beta_i(x) = \beta_i + \frac{A}{B}\frac{\theta}{T}\left(\frac{1}{\exp(\theta/T)-1}\right)\left[\left(x+\frac{\theta}{T}\right)^{1/2} + \exp\left(\frac{\theta}{T}\right)\left(x-\frac{\theta}{T}\right)^{1/2}\right]x^{3/2}, \quad T > \theta. \tag{17}$$

and

$$K(\beta_i,C\gamma_i) = \int_0^{\theta/T}\frac{x^3e^{-x}[x^2+\beta_i^1(x)]\,dx}{[x^2+\beta_i^1(x)]^2+C\gamma_i x^3} + \int_{\theta/T}^\infty \frac{x^3e^{-x}[x^2+\beta_i(x)]\,dx}{[x^2+\beta(x)]^2+C\gamma_i x^3}, \tag{18}$$

$$\beta_i = 6\mu_{Li}/\mu_{Ii}, \qquad \gamma_i = \tfrac{9}{16}\pi(\mu_i H/10^8)^2, \tag{19}$$

$$\mu_{L3} = AK/T^{3/2}, \tag{20}$$

$$\mu_{L2} = A/T^{3/2}, \tag{21}$$

where C is a constant that takes on the values a^2, $9a^2$, and $25a^2$ for $i=2$, and takes on the value 1 for $i=3$.

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TABLE I. Summary of the parameters used.

$p_3/p_2 = 0.046$
$\mu_{L3}/\mu_{L2} = 8.0 = K$
$\mu_{L3} = 3\times10^8/T^{3/2}$ cm^2/V sec
$m_3^* = 0.04m_e$
$m_2^* = 0.31m_e$
$(\epsilon_{op}/\epsilon_{ac})^2 = 3.8$

It is interesting that even with the inclusion of optical phonon scattering, the expression for the Hall coefficient factor is the same as when optical phonon scattering was ignored. Only the form for the transport integral changes, not the total expression for the Hall coefficient factor.

Using the expression given in Eq. (14), Eq. (12) can be written

$$1/R_H e = (p_2 + p_3)/\hat{\theta}. \tag{22}$$

Now all the parameters within the expression of the Hall coefficient factor $\hat{\theta}$ are known (see Tables I and II). Thus, if we know the concentration of the holes at any temperature, we can theoretically calculate the right-hand side of Eq. (22) and compare it with the experimentally determined value of the inverse of the Hall coefficient. In order to determine the total hole concentration, p_2+p_3, one must know the donor and acceptor concentrations because they are related by the charge neutrality equation:

$$p_2 + p_3 + N_D^+ = n + N_A^-. \tag{23}$$

For magnetic fields of the order of 1 kG or higher, the value of the Hall coefficient factor at temperatures below 70 °K can be taken as unity to a very good approximation, based upon a numerical computation. Using this approximation for Eq. (22), together with the neutrality equation [Eq. (23)] and the experimental Hall data in the low-temperature freezeout region, we determine the concentration of the acceptors and donors[5,12] and the energy level for the dominant impurity[12] for the crystals used in the present investigation, as shown in Table III. Since we are interested in calculating the Hall effect versus temperature, we must show the change of the band gap with temperature. For this we use the approximation[13]

$$E_g = E_{g0} - 4.4\times10^{-4}T, \tag{24}$$

where E_{g0} is taken to be equal to 0.8 eV at $T=0$.

TABLE II. Values of parameters determining the Hall coefficient factors for two types of energy surfaces.

	Spherical band model (Ref. 6)	Beer and Willardson model (Ref. 6)
a	1	0.935
b_3	0	0.085
b_5	0	0.035
α	1	0.96

TABLE III. Observed Hall mobilities and experimentally determined doping parameters.

	$\mu_{77°K}$ (cm^2/V sec)	N_A (cm^{-3})	N_D (cm^{-3})	E_A (meV)
High purity	45 000	5.6×10^{10}	3.0×10^{10}	12.8
In doped	44 000	1.6×10^{15}	9.6×10^{12}	11.3

III. EXPERIMENTAL PROCEDURE

The experimental measurements use the van der Pauw[14] configuration. The 10×10×1-mm samples have contacts < 1 mm^2 in the corners on the front side of the sample. The samples were cut from the the center of [100]-pulled Czochralski-grown boules up to 5 cm in diameter. The long dimensions of the sample were along the [010] and [001] directions. The resistivity variations at 77 °K across the sample were less than 3%. The rear side of the sample is mounted to a sapphire plate using a thin film of silicone heat-sinking compound. The sapphire plate, in turn, is mounted to a copper block containing two thermometers. The sample-block assembly is thermally choked from its support rod, which also supports a porous cylinder-heater assembly that contains the sample block. The purpose of this arrangement is to completely enclose the sample block with a temperature-feedback-controlled environment and to use cold helium gas to bring the sample block to a gradient-free equilibrium. One GaAs diode thermometer is used for temperature control, and the other is used for measuring temperature. In operation, a needle valve in the liquid-helium chamber is used to admit a constant flow of helium to the sample chamber. Two poppet valves are used to maintain a constant differential pressure of 1 psi across the needle valve. Once the needle valve has been properly adjusted for the coldest desired temperature, it remains fixed, and subsequent temperatures are achieved by increasing the heater current via the feedback controller. The sample current (typically 10 nA) is adjusted to keep the magnitude of the Hall voltage constant (typically 50 mV) in order to avoid errors caused by contact injection at high Hall voltages. A field-reversing electromagnet is used to obtain fields up to 10 kG. The Hall voltages ($V_{h\pm}$) are read for both polarities of the magnetic field to balance out offset voltages. The Hall coefficient is obtained from

$$R_H = \frac{E_x}{J_y B_z} = \frac{\bar{V}_h/L}{(I_s/Lt)B_0} = \frac{(V_{h+} - V_{h-})t}{2I_s B_0}, \tag{25}$$

where t is the sample thickness, I_s is the sample current, and B_0 is the magnetic field. The temperature voltage is read at the beginning and end of a set of readings (T_*, I_s, V_{h+}, V_{h-}, and T_*). These readings are typed into a programmable calculator that computes and records $1/R_H e$, T(°K), and $1/T$(°K^{-1}); the calculator also plots $\log_{10}(1/R_H e)$ vs $1/T$. The quantity $1/R_H e$ is approximately equal to the carrier concentration for low temperatures, and thus it is a useful experimental quantity to plot.

The accuracy of the Hall coefficient is about 5%—limited by the magnetic field and sample thickness uniformity. The precision for the Hall coefficient within a

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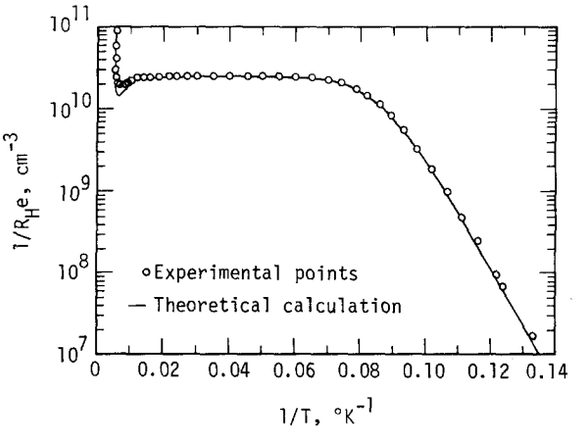


FIG. 1. Theoretical calculated inverse Hall coefficient as a function of the inverse temperature based upon the Beer and Willardson model (BW) compared with experimental data for a high-purity crystal at a magnetic field of 0.95 kG.

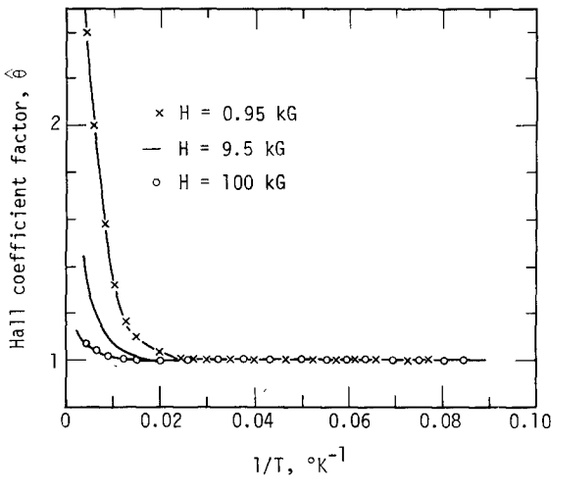


FIG. 3. Theoretical calculated Hall coefficient factor as a function of the inverse temperature based upon the BW model for three different values of the magnetic fields.

given sample run is limited by the sample current to about 0.1%. The accuracy of the temperature is about 0.05°K, and the precision within a given sample run is typically 0.02°K. The input impedance of the Hall voltage amplifier is 10^{10} Ω; thus, amplifier loading is negligible until the carrier concentration drops below 10^{6} cm^{-3}.

The preparation of reliable Ohmic contacts that are usable to about 5—6°K, where the amplifier input impedance becomes the limiting factor, has been achieved using a simple flux-free process. The p-type sample is etched in 3:1 HNO_3:HF, followed by a DI water rinse. A small spot of liquid metal, 1:4 In:Ga, is placed on the corners, and the sample is then heated on a hot plate in N_2 to 350°C for 5 min and cooled slowly. The excess liquid metal is carefully removed with cotton swabs, and indium solder contacts are made with an iron at 200°C.

IV. EXPERIMENTAL RESULTS

Figures 1—3 show the experimental results and the theoretical calculations for the inverse of the Hall coefficient, actually $1/R_He$, and for the Hall coefficient factor as a function of the inverse temperature for a high-purity crystal (impurity concentration in the 10^{10}-cm^{-3} range). The quantity $1/R_He$ is plotted, since it is obtained directly during the experiment and since it is approximately equal to the hole concentration over the lower exhaustion and freezeout range. The calculations shown in these figures were based upon the Beer-Willardson (BW) model. As one can see from these figures, the shape of the theoretical curve agreed very well with the experimental curves, especially at the magnetic field of 9.5 kG.

The Hall coefficient factor $\hat{\theta}$ as a function of the inverse temperature was plotted in Fig. 3 for three different values of the magnetic fields. $\hat{\theta}$ does not change very much until it reaches temperatures of 25°K or higher for a magnetic field of 0.95 kG, and it remains unchanged until the temperature reaches about 50°K or higher for a magnetic field of 9.5 kG. After about 80°K

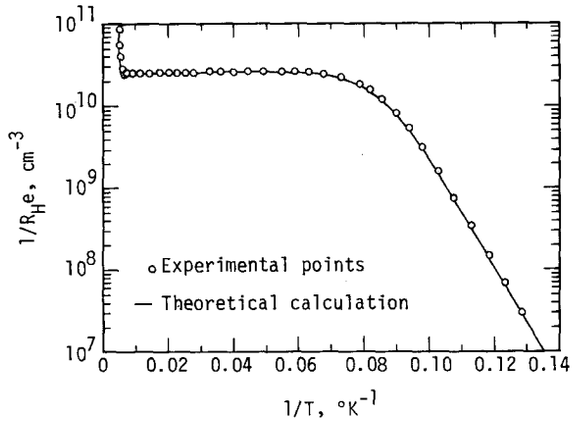


FIG. 2. Theoretical calculated inverse Hall coefficient as a function of the inverse temperature based upon the BW model compared with experimental data for a magnetic field of 9.5 kG.

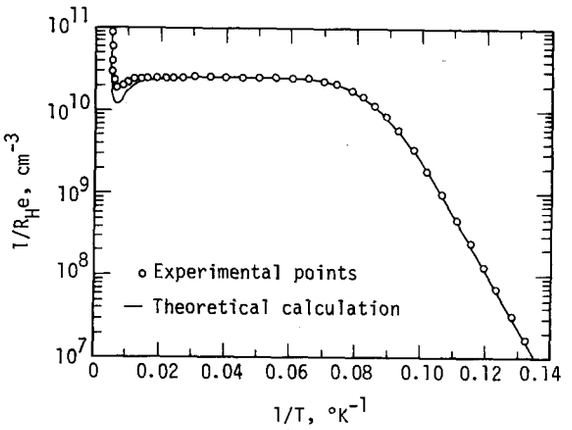


FIG. 4. Theoretical calculated inverse Hall coefficient as a function of the inverse temperature based upon a spherical-band model for both light and heavy holes compared with experimental data for a high-purity crystal at a magnetic field of 0.95 kG.

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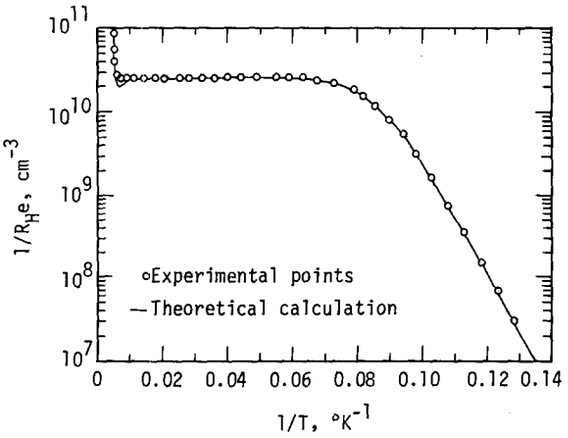


FIG. 5. Theoretical calculated inverse Hall coefficient as a function of the inverse temperature based upon a spherical-band model for both light and heavy holes compared with experimental data for a high-purity crystal at a magnetic field of 9.5 kG.

all the Hall coefficient factors increase rapidly with temperature. It is the rapid increase in the value of $\hat{\theta}$ in the 80—200°K temperature range that drives the inverse Hall coefficient to a minimum value before the onset of intrinsic conduction.

Figures 4 and 5 compare the theoretical calculation with observed data for the inverse Hall coefficient based on the spherical-band approximation for both the light and heavy holes. As can be seen from these figures, the simple spherical-band approximation does not give as good agreement as the BW model does.

Figure 6 shows how the Hall coefficient factor varies with the inverse of the temperature, based upon two different models. As can be seen from these figures, $\hat{\theta}$

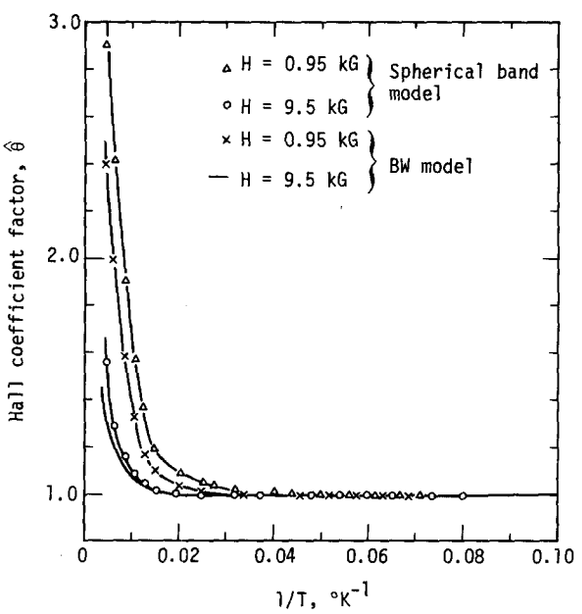


FIG. 6. Theoretical calculated Hall coefficient factor as a function of the inverse temperature based upon the BW model and the spherical-band model for two different values of the magnetic fields.

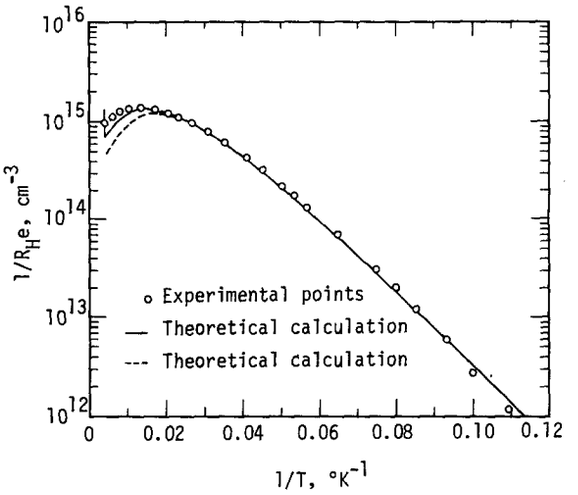


FIG. 7. Theoretical calculated inverse Hall coefficient of an indium-doped sample as a function of the inverse temperature compared with experimental data for a magnetic field of 1.2 kG. The dashed curve applies to the spherical band model and the solid curve applies to the BW model.

based on the BW model did not change as rapidly as $\hat{\theta}$ based on the spherical-band model.

Figures 7 and 8 show the inverse Hall coefficient for a highly doped p-type crystal. The crystal used in these measurements was intentionally doped with indium. Both the spherical-band and the BW models were used in the calculation for Figs. 7 and 8. Again, one can see from these figures that the theoretical calculation agrees quite well with the experimental result for the inverse Hall coefficient.

We do not show the Hall coefficient factor as a function of the inverse temperature for the intentionally doped sample because the difference between the $\hat{\theta}$ of the unintentionally doped sample and that of the inten-

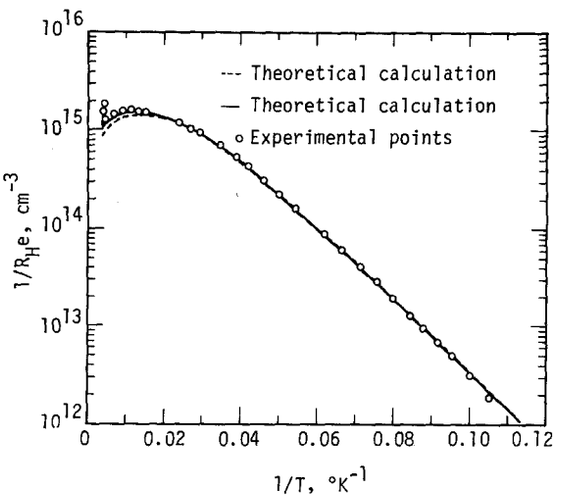


FIG. 8. Theoretical calculated inverse Hall coefficient of an indium-doped sample as a function of the inverse temperature for two different theoretical models compared with experimental data for a magnetic field of 9.5 kG. The dashed curve applies to the spherical-band model and the solid curve applies to the BW model.

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tionally doped samples is graphically insignificant. In other words, within the doping range investigated, changes in the impurity concentrations had almost no influence on $\hat{\theta}$.

V. CONCLUSION

In the present paper we have modified the expression for the Hall coefficient factor that was originally derived by Beer and Willardson for p-type germanium by incorporating the optical phonon scattering mechanism into the equation. The modified expression has exactly the same form as the original one; only the transport integrals change [see Eqs. (15)—(21)]. In evaluating the Hall coefficient factor for temperatures higher than 70°K, one must use the modified transport integrals, because above this temperature optical phonon interaction is very important.

The Hall effect in two samples with two different impurity concentrations was investigated as a function of temperature and as a function of magnetic field. The main purpose of our Hall effect study is to try to understand the behavior of the inverse of the Hall coefficient in the temperature range from 70°K to the temperature at which the onset of intrinsic conduction occurs. The behavior of the Hall effect in this temperature range has never been studied theoretically. It must be emphasized that it is not easy to obtain a crystal in which the inverse of the Hall coefficient shows a minimum as the temperature increases before the onset of the intrinsic conduction. The temperature at which the minimum occurs is quite high. If the crystal has some deep impurity or defect centers, the centers can detrap carriers, which then, in turn, obscures the effect.

Extensive theoretical calculations were carried out in analyzing the behavior of the Hall-effect curve. By comparing the theoretical calculation with the experimental results, the theory shows excellent agreement with the experimental data.

It is important to point out that all the parameters used in our calculation were determined from previous experiments. Therefore, no adjustable parameters were employed in our calculation.

Although our theory was based upon a single shallow-acceptor level, this simplicity should not invalidate our calculation. This can be seen as follows. In the temperature range from about 70°K up to the onset of intrinsic conduction, all shallow levels of the acceptors were ionized. Hence, in this temperature range, only the free carriers, the magnetic field, and the band structure influence the shape of the Hall curve.

At impurity concentrations around 10^{10} cm^{-3}, the existing impurity scattering theory is no longer valid for the mobility calculation. However, because the concentration was so low in this sample, the phonon scattering is still the most important scattering mechanism. Therefore, the use of the formula of Eq. (4) will not alter the values of our calculation.

ACKNOWLEDGMENT

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